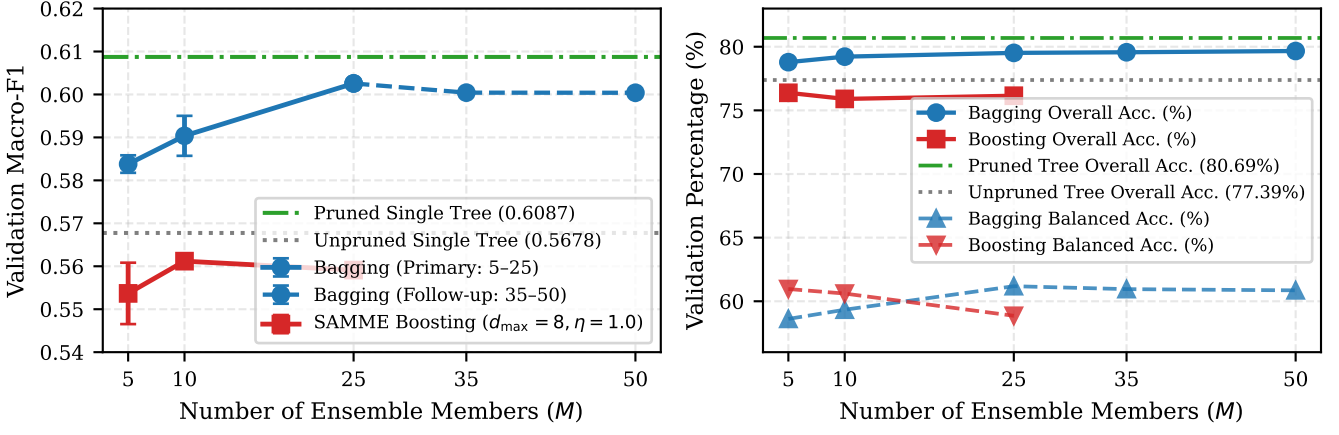
(a) Validation Macro-F1 vs. Ensemble Size
Validation Macro-F1
0.62
0.61
0.60
0.59
0.58
0.57
0.56
0.55
0.54
5
10
25
35
50
Number of Ensemble Members (M)
Pruned Single Tree (0.6087)
Unpruned Single Tree (0.5678)
Bagging (Primary: 5–25)
Bagging (Follow-up: 35–50)
SAMME Boosting ($d_{\max} = 8, \eta = 1.0$)
(b) Validation Accuracy & Balanced Accuracy
Validation Percentage (%)
80
75
70
65
60
5
10
25
35
50
Number of Ensemble Members (M)
Bagging Overall Acc. (%)
Boosting Overall Acc. (%)
Pruned Tree Overall Acc. (80.69%)
Unpruned Tree Overall Acc. (77.39%)
Bagging Balanced Acc. (%)
Boosting Balanced Acc. (%)